# Normalization Protocol for Electricity Theft Detection Datasets

SGCC, CER, London LCL and ETD2022

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**Abstract.** This protocol defines the normalization of the common daily record for offline batch classification of electricity theft: five partitions in which every meter appears in exactly one of the training, validation and test sets, a scaler fitted on SGCC training consumers only, a relative profile based on each meter's own mean, and nonoverlapping windows of 30 consecutive days. The final form stores every dataset as the same record of 30 daily values. The protocol also states the leakage controls, the validity boundaries, the verification procedure and the outputs of the two stages.

**Keywords:** normalization; data leakage; domain shift; daily load profiles; electricity theft detection

## 1 Purpose and Scope

The protocol applies to the common daily table produced by the preprocessing stage. The audited kWh values remain unchanged, and every normalized value is a new column. The work is an offline batch preparation. Stage 2 holds the partitions of the meters, the scaler parameters and the mean of each meter. Stage 3 holds the final common form, with one record of 30 daily values per window and the same columns for all four datasets.

## 2 Partitions by Meter

Five partitions are generated with the seeds 20261021 to 20261025. In each partition every meter is assigned to exactly one of the training, validation and test sets, and the series of a meter is never divided over time. SGCC is stratified by its consumer label. ETD2022 is the single outside check of the research plan, so every block is test in every seed. It draws no random numbers, which leaves the SGCC, CER and London LCL assignments unchanged by its presence.

**Table 1.** Partition ratios.

| Domain | Training / validation / test | Stratification | Basis of the ratio |
|---|---|---|---|
| **SGCC** | 70 / 10 / 20 | Consumer label | Research plan |
| **CER** | 60 / 10 / 30 | None, labels unknown | Research plan target split |
| **London LCL** | 60 / 10 / 30 | None, labels unknown | Assumed, because the plan gives no ratio for this domain |
| **ETD2022** | 0 / 0 / 100 | Not applicable | Research plan, single outside check |

## 3 Absolute Scale

The absolute scale is the literal minimum and maximum scaling specified by the research plan.

*scaled value = (daily kWh minus minimum) divided by (maximum minus minimum) (1)*

The minimum and the maximum are computed from the eligible daily readings of the SGCC training consumers of the seed. They are fixed once and applied without change to SGCC validation and test consumers and to CER, London LCL and ETD2022. No clipping is applied, so a scaled value outside 0 to 1 is possible. The minimum is 0.0 for every seed, and Table 2 gives the remaining parameters.

**Table 2.** Scaler parameters fitted on SGCC training consumers.

| Seed | Consumers | Eligible days | Maximum (kWh per day) | 99th percentile (kWh per day) | 99th percentile after scaling |
|---|---|---|---|---|---|
| 20261021 | 29,656 | 22,986,787 | 800,003.31 | 68.55 | 0.000086 |
| 20261022 | 29,656 | 23,039,718 | 514,991.78 | 68.59 | 0.000133 |
| 20261023 | 29,657 | 23,029,948 | 800,003.31 | 72.88 | 0.000091 |
| 20261024 | 29,656 | 23,009,542 | 653,502.81 | 71.31 | 0.000109 |
| 20261025 | 29,655 | 23,030,871 | 800,003.31 | 70.50 | 0.000088 |

## 4 Relative Profile

The relative profile expresses each daily reading as a multiple of the meter's own average, as required by the research plan.

*relative value = daily kWh divided by the mean of the eligible daily kWh of the same meter (2)*

The mean uses the whole series of the meter and no label. The profile is a batch transform and is not causal. A meter whose mean is zero has no defined profile, so its relative values stay empty, its windows carry empty relative arrays and `relative_defined` is FALSE. Physical kWh and the absolute scale remain available for such meters.

## 5 Equal Length Windows

The research plan requires sequences of equal length but gives no number. A window of 30 consecutive eligible days with a stride of 30 days, so that windows do not overlap, is adopted as a documented choice. A window never crosses an ineligible day, a long gap or a meter boundary, and no padding is added. Spans shorter than 30 days and remainders are not used, and the number of contiguous runs and of eligible days left unused is reported in the Summary Report. A block of ETD2022 with 365 relative days gives 12 windows and leaves 5 days unused. Each window carries an identifier, its domain, its meter and its start day and end day.

## 6 Final Common Form

The final form stores one row per window, with the same columns for all four datasets (Table 3). Each row holds three arrays of 30 values for the same days: physical kWh, the absolute scale of the seed and the relative profile. It also holds the split of the seed, the label and the calendar basis. The label is 1 when at least half of the 30 days carry a theft label and 0 otherwise, and it is empty where labels are unknown. For SGCC this equals the consumer flag, because every day of a consumer carries it. For ETD2022 it uses the day labels, and `label_fraction` keeps the exact share so that another threshold can be applied. One table is written per seed, because the split and the absolute scale depend on the seed. In the Excel workbooks each array is written as 30 columns.

**Table 3.** Columns of the final window tables.

| Column | Type | Meaning |
|---|---|---|
| `window_id` | Text | Unique identifier of the window |
| `domain, meter_id` | Text | Dataset and meter, consumer or inferred block |
| `start_day, end_day` | Date | First and last of the 30 days |
| `calendar_basis` | Text | source or nominal |

**Table 3 (continued).** Columns of the final window tables.

| Column | Type | Meaning |
|---|---|---|
| `split` | Text | train, validation or test for the seed of the table |
| `label` | Number | 1 when at least half of the days carry a theft label, 0 otherwise, empty when unknown |
| `label_fraction` | Number | Share of the 30 days that carry a theft label, empty when unknown |
| `label_scope` | Text | consumer, day or none |
| `label_is_synthetic` | TRUE or FALSE | TRUE for ETD2022 |
| `relative_defined` | TRUE or FALSE | FALSE when the meter mean is zero |
| `imputed_days` | Number | Days of the window with at least one interpolated interval |
| `has_high_spike` | TRUE or FALSE | TRUE when a day of the window is flagged as an outlier |
| `kwh_d01 to kwh_d30` | 30 numbers | Physical daily energy in kWh, one column per day |
| `kwh_absolute_minmax_d01 to d30` | 30 numbers | Absolute scale of the seed, equation 1 |
| `kwh_relative_meter_mean_d01 to d30` | 30 numbers | Relative profile, equation 2, empty when undefined |

## 7 Meter Attributes

The tariff group, ACORN code and ACORN group of the 5,566 London households are columns of the London LCL daily table of stage 1. The building archetype and the source rows of the 64 ETD2022 blocks are columns of the ETD2022 daily table. Both are keyed by `domain` and `meter_id`. These fields are covariates. They are not theft labels and are not encoded numerically.

## 8 Validity Boundaries and Leakage Controls

**Table 4.** Boundaries that govern the use of the outputs.

| Topic | Boundary |
|---|---|
| **Leakage** | Scalers, imputers and decision thresholds must be fitted on SGCC training consumers only. No imputer or threshold has been fitted at this stage. |
| **Offline use** | Interpolation uses values on both sides of a gap, outlier flags use the full series of a meter, and the relative profile uses its overall mean. The views suit offline batch classification. For causal deployment they must be rebuilt from past information only. |
| **Calendar** | SGCC, CER and London LCL do not share a calendar period. SGCC and London LCL have 58 days in common, from 1 January to 27 February 2014, and CER has none in common with either. ETD2022 days are nominal and correspond to no real period. |
| **Daylight saving** | CER days with 46 or 50 slots keep their observed totals. A sensitivity analysis may exclude or adjust them using the recorded slot counts. |
| **Scale compression** | The literal scaling of Equation 1 places typical readings of SGCC, CER and London LCL far below 0.001, because the SGCC training maximum is very large. ETD2022 buildings consume far more, so their scaled median is about 0.001 to 0.002. Physical kWh is retained, and a robust scale fitted on SGCC training consumers only should be assessed at modelling time. |

**Table 4 (continued).** Boundaries that govern the use of the outputs.

| Topic | Boundary |
|---|---|
| **Labels** | SGCC labels describe consumers and not dated events. CER and London LCL labels are unknown. ETD2022 labels are synthetic and hourly, so 43.8 percent of its days and 31.0 percent of its windows are labelled theft, against 8.5 percent of SGCC consumers. Synthetic attack injection for CER, the planned 8.5 percent prevalence and any target label budget are later experimental steps and must be logged separately. |
| **ETD2022 identity and calendar** | Blocks are inferred sequences and not verified meters, the calendar is nominal, and energy is estimated from power. Weekday and month features are not meaningful for ETD2022. Its window label depends on the threshold of one half, and label_fraction allows others. |
| **Departure from the plan** | The research plan proposes zero filling of long gaps and a three sigma cap on outliers. The supervisor's instruction takes precedence and requires exclusion and logging of long gaps and a flag at five times the 99th percentile. This protocol follows the instruction. |
| **Metadata** | ACORN assignments are verified only against the Kaggle copy. The customer segment of CER (residential, SME or other) is given in the publisher's allocation file and has not been joined to the outputs. |

## 9 Verification Procedure

Every output is verified before use. Each meter must appear in exactly one partition for each seed, in the counts given by Table 1. The scaled SGCC training range must be 0 to 1 for each seed, both formulas must be recomputed on every value, and the mean of the relative profile of every meter must equal 1. Every window must span exactly 30 consecutive days with a unique identifier, and a selection of windows must agree with the source days. The values of the final windows must equal the normalized daily values exactly for a selection that includes every ETD2022 window. The results for SGCC, CER and London LCL must equal those of an independent normalization run of the same data. The results are reported in the Summary Report.

## 10 Outputs of Stages 2 and 3

All paths are relative to the collection folder CER4. An Excel sheet holds at most 1,048,576 rows. A table that fits one sheet is delivered in one workbook. A larger table is delivered in consecutive numbered workbooks of at most 1,000,000 rows, each with its own header, and the parts together hold every row of the table.

The final window tables hold the normalized values of every day that forms a window. The normalized daily tables of 39,712,070 rows for each seed and the window index are not delivered as separate sets of workbooks.

**Table 5.** Workbooks of stages 2 and 3.

| Workbook | Content | Rows of the table | Form |
|---|---|---|---|
| `2_normalization/split_manifest_five_seeds.xlsx` | Partition of every meter for the five seeds | 272,160 | Complete |
| `2_normalization/scaler_parameters.xlsx` | Scaler parameters of the five seeds | 5 | Complete |
| `2_normalization/meter_summary.xlsx` | Eligible days and mean of every meter | 54,432 | Complete |
| `3_final_common_normalized/final_windows_30d_seed_SEED_part1_of_2.xlsx and part2_of_2` | Final windows of all four datasets. Ten workbooks, two for each SEED from 20261021 to 20261025 | 1,275,674 for each seed | Complete, 2 workbooks per seed |

## References

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[3] UK Power Networks. SmartMeter Energy Consumption Data in London Households (Low Carbon London). London Datastore.

[4] Kaggle dataset Smart meters in London (user jeanmidev), source of the ACORN household attributes.

[5] ETD2022 electricity theft dataset, version 3. Mendeley Data, doi:10.17632/c3c7329tjj.3.